EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)

Topic	**D: Further Algebra and Functions**
Sub topic	**D7: Evaluate Limits Using L'Hôpital's Rule**

Mark Scheme

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To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

$\sqrt{4+x} = 2\left(1+\frac{x}{4}\right)^{\frac{1}{2}} = 2\left[1 + \frac{1}{2}\left(\frac{x}{4}\right) + O(x^2)\right]$>

Attempt to use binomial theorem OE The notation $O(x^n)$ can be replaced by a term of the form kx^n

M1

$\left[\frac{\sqrt{4+x}-2}{x+x^2}\right] = \left[\frac{\frac{x}{4}+O(x^2)}{x+x^2}\right] = \left[\frac{\frac{1}{4}+O(x)}{1+x}\right]$

Division by x stage before taking the limit

M1

$\lim_{x\to 0}\left[\frac{\sqrt{4+x}-2}{x+x^2}\right] = \frac{1}{4}$

CSO NMS 0 / 3

A1

[3]

Examiner reports

Q1.

This question, which tested limits, proved to be the least well answered question on the paper. Usually candidates have been guided into the method on previous papers and, without the guidance of earlier parts to the question, the majority of candidates just presented solutions based on replacing x by 0 and claiming that 0 / 0 was some definite value. The most successful candidates started by finding the series expansion of $\sqrt{4+x}$ and then divided both the numerator and denominator by x before taking the limit. A small minority of candidates started by multiplying the numerator and denominator by $\sqrt{4+x}$ + 2 and usually went on to show that the value of the limit was 1/4. Answers with no method seen were awarded no credit.